



Optimal Design of Digital IIR Filters Using Robust Group Search Optimization

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Abstract

This paper presents a novel approach to designing the optimal lowest-order serial digital Infinite Impulse Response (IIR) filters using robust group search optimization (R-GSO), where the structures of IIR filters are specified in advance. The R-GSO algorithm integrates the GSO algorithm and the Taguchi method. The number of producers of the original GSO algorithm is fixed to one, which makes it easy to fall into the local optimal solution. The R-GSO algorithm selects three producers and applies the Taguchi method to determine the optimal producer for each iteration to enhance the robustness of the search process and the performance of the global search optima. The Taguchi method not only considers the solutions with the highest fitness but also evaluates the robustness of the solutions. Digital IIR filter design is a complex, multi-parameter, multi-objective optimization problem, and the objectives are often at odds. The performance of a digital IIR filter can be evaluated by the Lp-norm error, the ripple size of the passband, and the ripple size of the stopband. Thus, this paper considers the L1-norm, L2-norm, and the ripple size of the passband and the stopband to search for a flatter and closer to the target of the lowest order of the digital IIR filter. From the experimental results, the R-GSO algorithm effectively optimizes the digital low-pass (LP), high-pass (HP), band-pass (BP), and band-stop (BS) filters. It outperforms the hybrid Taguchi genetic algorithm (HTGA) and original GSO algorithm. The RGSO algorithm also has fewer function evaluations than HTGA, making it possible to optimize these four filters more efficiently and apply them to real-world engineering applications where high-performance digital filters are required.

Keywords Digital IIR filter · Multi-objective optimization · Robust group search optimization · Taguchi method

Extended author information available on the last page of the article

1 Introduction

A digital filter is a widely used tool in digital signal processing (DSP), primarily for processing, improving, or analyzing signals. Digital filters manipulate the input digital signal through specific mathematical methods to remove noise, extract proper signals, or perform signal frequency analysis. Digital filters are broadly categorized into finite impulse response (FIR) and infinite impulse response (IIR) filters.

Two principal methodologies are commonly employed for the design of digital IIR filters: the transformation approach and the optimization approach. Classical techniques such as Butterworth, Chebyshev, or Elliptical are used to create an analog prototype filter in traditional IIR filter design [2, 18]. The bilinear transformation [23] approach involves transforming an analog filter design into a digital filter, which effectively maps the frequency response of the analog filter into the digital domain while preserving the stability of the filter. However, it can introduce warping effects at high frequencies, which may require compensation. The optimization approach uses numerical techniques to optimize the filter coefficients directly in the digital domain. The p-error, mean-square-error, and desired ripple of both passband and stopband are criteria to measure the performance of the designed digital IIR filters [5, 6, 13, 15, 17, 25, 27].

The digital IIR filter design is a multi-objective optimization problem with multiple local optima. Therefore, the minimum order of the filter, the stability of the filter, and fulfillment of the tolerance settings constraints must be strictly imposed when designing a digital IIR filter. The optimization procedure determines the fulfillment of the tolerance settings by minimizing the ripple magnitudes in both the passband and stopband [26]. These constraints take much work to optimize.

In recent years, some researchers have used optimization methods to solve digital IIR filter design problems [2, 3, 7–9, 14, 16, 20–22, 27]. Chauhan et al. proposed a diversity-driven multi-parent evolutionary algorithm (DDMPEA-ANUM) to design digital IIR filters, considering the magnitude response errors in the passband and stopband, as well as the linear phase response errors as fitness functions, optimized within a multi-objective framework [3]. Liang et al. proposed an improved slime mould algorithm (ESMA) for digital IIR filter design, incorporating chaotic initialization, orthogonal learning, and boundary reset strategies [16]. Ekinici et al. proposed an artificial hummingbird algorithm (AHA) and atom search optimization (ASO) for optimizing digital IIR filter models to identify optimal coefficients [7, 8]. Wu applied the non-dominated sorting genetic algorithm II (NSGA-II) to design digital IIR filters, optimizing the minimum passband ripple and the maximum stopband attenuation [28]. Abualigah et al. proposed a modified atom search optimizer (mASO), which combines an adaptive gbest-guided mechanism to overcome weaknesses in the standard atom search optimizer to design digital IIR filters [1]. In summary, designing digital IIR filters is a complex task involving multi-objective parameters, and targets are often at odds. A significant challenge arises with bio-based methods potentially getting stuck in the local optima of the multi-objective problem.

The serial digital IIR filters include low-pass (LP) filter, high-pass (HP) filter, band-pass (BP) filter, and band-stop (BS) filter. The Lp-norm error [26] and the ripple sizes in the passband and stopband [17] can evaluate the performance of a digital IIR

filter. Under the situation of the structures of IIR filters are specified in advance, from the presented results of designing the optimal digital IIR filters in the literature, it can be seen that the design cases studied by Tsai et al. [27], includes LP, HP, BP and BS filters as well as the design results proposed by Tsai et al. [27] are better than those given in the literature.

Therefore, under the situation where the structures of IIR filters are specified in advance, this paper applies robust group search optimization (R-GSO) [29] to design the optimal lowest-order serial digital IIR filters and compares them with the design results given by Tsai et al. [27]. The R-GSO method combines the original GSO [10] and Taguchi methods [19]. The original GSO selects the best producer in the group as its exploration search algorithm, and then, through the producer-scrounger (PS) mechanism, the other individuals in the group will approach the best position. However, only one best producer is selected to guide another scrounger to the best position; it is often trapped in a local optimal solution and cannot do a perfect exploration search [11]. The R-GSO selects three producers and combines them with Taguchi methods to determine the optimal producer in each iteration to improve the global search and stabilization performance.

In a previous study [29], the performance of the R-GSO algorithm is compared with the original GSO algorithm [10] on three unimodal, three multimodal, and four limited multimodal benchmark functions. While the R-GSO algorithm only outperformed the original GSO algorithm on two of the three unimodal functions, it demonstrated greater robustness overall. Furthermore, the R-GSO algorithm exhibited superior performance and robustness compared to GSO algorithm in multimodal and limited multimodal functions.

The organization of this paper is as follows: Section 2 outlines the problem statement and application methods. Section 3 explores the implementation of R-GSO algorithm in designing digital IIR filters, comparing its performance with that of other researchers for LP, HP, BP, and BS filters. Section 4 summarizes the findings and concludes the study.

2 Problem Statement and Application Methods

2.1 Problem Statement

A common method for implementing IIR filters is to cascade multiple first-order and second-order sections. Let $H(\omega, x)$ be the transfer function of the digital filter, where x represents the coefficient of the poles and zeros of the filter; $|H(\omega, x)|$ can be expressed as the size of $H(\omega, x)$. For any serial digital IIR filter, the transfer function can be described as follows [24]:

$$H(\omega, x) = \alpha \gamma \left(\prod_{i=1}^M \frac{1 + a_i e^{-j\omega}}{1 + b_i e^{-j\omega}} \right) \left(\prod_{k=1}^N \frac{1 + c_{1k} e^{-j\omega} + c_{2k} e^{-2j\omega}}{(1 + d_{1k} e^{-j\omega} + d_{2k} e^{-2j\omega})} \right) \quad (1)$$

where $x = (a_1, b_1, \dots, a_{1M}, b_{1M}, c_{11}, c_{22}, d_{11}, d_{21}, \dots, c_{1N}, c_{2N}, d_{1N}, d_{2N}, \gamma)$ represents the coefficient of the filter's poles and zeros, and α is a positive normalizing constant with an adjusted the maximum value of $|H(\omega, x)|$ to 1.

Equation (1) is the result if the numerator and denominator of the IIR consideration filter are the same. However, with appropriate adjustments and assumptions, it can also be applied to the case where the numerator and denominator are different.

The performance of a digital IIR filter can be evaluated by the Lp -norm error [26], the ripple size of the passband, and the ripple size of the stopband [17]. Therefore, when designing an optimal digital IIR filter, the evaluation function can consider the error of the Lp -norm as well as the ripple size of the passband and the stopband. In general, using equidistant fixed grid points can calculate the error of Lp -norm, as described in the following equation [26]:

$$E_M^P(x) = \left\{ \sum_{i=1}^K |T_M(\omega_i) - |H(\omega_i, x)||^p \right\}^{\frac{1}{p}}, \omega_i \in \Omega, \text{ and } p = 1 \text{ and } 2 \quad (2)$$

where ω_i is the i -th discrete frequency point, Ω represents the combined frequency range of the passband and stopband that the design targets; K denotes the total number of discrete frequency points equidistant within the combined frequency range of the passband and stopband; p is the order of the Lp -norm, $p = 1$ and 2 are the most popular in filter design problems; $T_M(\omega_i)$ is the target amplitude response at ω_i and is set as follows:

$$T_M(\omega_i) = \begin{cases} 1, & \text{for } \omega_i \in \text{passband.} \\ 0, & \text{for } \omega_i \in \text{stopband.} \end{cases} \quad (3)$$

If $p = 1$, then $E_M^1(x)$ represents the absolute value of the error between the amplitude response and the target value (i.e., L_1 -norm); if $p = 2$, then $E_M^2(x)$ represents the squared error between the amplitude response and the target value (i.e., L_2 -norm); Let $\delta_1(x)$ and $\delta_2(x)$ denote the ripple sizes of the passband and stopband, respectively. The $\delta_1(x)$ and $\delta_2(x)$ are defined as follows:

$$\delta_1(x) = \max_{\omega_i} \{|H(\omega_i, x)|\} - \min_{\omega_i} \{|H(\omega_i, x)|\}, \text{ for } \omega_i \in \text{passband.} \quad (4)$$

and

$$\delta_2(x) = \max_{\omega_i} \{|H(\omega_i, x)|\}, \text{ for } \omega_i \in \text{stopband.} \quad (5)$$

In the multi-objective optimization of digital IIR filter design, the design of the evaluation function determines the generation of a non-inferior or optimal tradeoff point. This paper considers four criteria the L_1 -norm, L_2 -norm, and the ripple size of the passband and the stopband to search for a flatter and closer to the target of the lowest order of the digital IIR filter. In most situations, these four criteria are often in conflict with each other. The optimal tradeoff point can generate by solving:

$$\min_x f(x) = \min_x v_1 E_M^1(x) + v_2 E_M^2(x) + v_3 \delta_1(x) + v_4 \delta_2(x) \quad (6)$$

The constraints on the coefficient stability of a serial digital IIR filter are as follows [12]:

$$\begin{aligned} 1 + b_i &\geq 0, \\ 1 - b_i &\geq 0, \\ 1 - d_{2k} &\geq 0, \\ 1 + d_{1k} + d_{2k} &\geq 0, \\ 1 - d_{1k} + d_{2k} &\geq 0 \end{aligned} \quad (7)$$

where $f(x)$ is the objective function, $i = 1, 2, \dots, M$; $k = 1, 2, \dots, N$; v_1, v_2, v_3 , and v_4 are the weights between the objectives, and their values are non-negative real numbers. Different values of v_1, v_2, v_3 , and v_4 represent different weights on $E_M^1(x)$, $E_M^2(x)$, $\delta_1(x)$, $\delta_2(x)$. The choice of these weights affects the optimal trade-off point. This weight alters the optimal tradeoff point. Cortelazzo and Lightner have proposed the technique for selecting appropriate weight values [5]. In addition, the stability constraints given in equ. 7 are essential when designing serial-optimized digital IIR filters. Therefore, the poles of the digital filters must be located in the unit circle.

The serial digital IIR filters include LP filter, HP filter, BP filter, and BS filter. The design constraints of the above four types of filters in terms of the frequency range of the passband and stopband are described in Table 1.

2.2 Robust-GSO Algorithm

The Group Search Optimization (GSO) algorithm relies on random searchers (Rangers) to perform exploitation search operations. In original GSO implementations, a single best-fit producer (Producer) is typically selected to conduct exploration searches. Through the producer-scrouter (PS) mechanism, other group members are guided toward the position with the highest fitness. However, selecting only one producer often leads the algorithm to get trapped in local optima, limiting its ability to explore more broadly. To avoid trapping in local optima, this paper utilizes the R-GSO [29], which selects a fixed number of multiple producers combined with the Taguchi method to determine the optimal producer in each iteration. The Taguchi method is a design of experiments based on statistical theory. The Taguchi method can obtain robustness by planning the minimum number of experiments among the selected factors using an orthogonal table and adding factors to the experiments that

Table 1 Design conditions for LP, HP, BP and BS filters

Filter type	Passband	Stopband	Maximum value of $ H(\omega, x) $
LP	$0 \leq \omega \leq 0.2\pi$	$0.3\pi \leq \omega \leq \pi$	1
HP	$0.8\pi \leq \omega \leq \pi$	$0 \leq \omega \leq 0.7\pi$	1
BP	$0.4\pi \leq \omega \leq 0.6\pi$	$0 \leq \omega \leq 0.25\pi$ $0.75\pi \leq \omega \leq \pi$	1
BS	$0 \leq \omega \leq 0.25\pi$ $0.75\pi \leq \omega \leq \pi$	$0.4\pi \leq \omega \leq 0.6\pi$	1

are uncontrollable in the environment where the system is being used, but can affect the quality characteristics [19]. Therefore, the Taguchi method enables efficient exploration of multiple producer combinations with a minimum number of experiments. The Taguchi method not only considers the solutions with the highest fitness but also evaluates the stability of the solutions, ensuring that the selected producer can maintain good performance across multiple experiments. In this paper, the Taguchi method is used to enhance the global search capability of the GSO algorithm. This approach not only enhances the exploration search mechanism but also significantly improves the robustness of the solutions, effectively expanding the search range and enhancing the algorithm's global optimization capabilities.

In the initial setup of the R-GSO algorithm, a parameter X_p representing the number of producers is introduced, with a recommended range of $X_p \in [1, 10]$. During each iteration, the algorithm selects the individuals with the best and second-best fitness scores as producers, labeled as X_{p1} and X_{p2} , respectively. To ensure diversity, a third producer, X_{p3} , is selected from the remaining group members. These three producers are used to set the levels of variables in an orthogonal experimental design. Therefore, three crossover types are selected. Each iteration involves setting X_{p1} and X_{p2} as the first and second levels, respectively, and similarly for combinations of X_{p1} , X_{p2} , and X_{p3} . An orthogonal table $L_m(2^{m-1})$ is selected, and three experiments are conducted for different level combinations of X_p , where m denotes the number of rows of the orthogonal experiment; $m - 1$ denotes the number of columns of configurable decision parameters. When the Taguchi method is completed, three new producers are generated, X_{np1} , X_{np2} , and X_{np3} , respectively. These new producers determine the optimal producer X_{op} for each iteration, enhancing the robustness of the search process and the performance of the search global optima.

For example, each producer has seven dimensions and two levels. The orthogonal table $L_8(2^7)$ is used, as shown in Tables 2, 3, and 4. Upon completion of the orthogonal experiment, each row combination yields three results. The robust response value equ. 8 is used to find the level that contributed more to each factor, and ensures robustness in the search for the optimal value [4]. If $E_{f1} > E_{f2}$, the optimal level of factor f is the first level. Otherwise, it is the second level.

$$E_{fj} = -10 \cdot \log \left(\left(J_{i(\text{avg})}^2 / J_{\text{Mavg}}^2 \right) + \left(J_{i(\text{std})}^2 / J_{\text{Mstd}}^2 \right) \right), j = 1, 2 \quad (8)$$

Table 2 Example of crossover performed using $L_8(2^7)$ orthogonal table for X_{p1} and X_{p2}

Factor							
Exp.	A	B	C	D	E	F	G
1	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}
2	X_{p1}	X_{p1}	X_{p1}	X_{p2}	X_{p2}	X_{p2}	X_{p2}
3	X_{p1}	X_{p2}	X_{p2}	X_{p1}	X_{p1}	X_{p2}	X_{p2}
4	X_{p1}	X_{p2}	X_{p2}	X_{p2}	X_{p2}	X_{p1}	X_{p1}
5	X_{p2}	X_{p1}	X_{p2}	X_{p1}	X_{p2}	X_{p1}	X_{p2}
6	X_{p2}	X_{p1}	X_{p2}	X_{p2}	X_{p1}	X_{p2}	X_{p1}
7	X_{p2}	X_{p2}	X_{p1}	X_{p1}	X_{p2}	X_{p2}	X_{p1}
8	X_{p2}	X_{p2}	X_{p1}	X_{p2}	X_{p1}	X_{p1}	X_{p2}

Table 3 Example of crossover performed using $L_8(2^7)$ orthogonal table for X_{p1} and X_{p3}

Factor							
Exp.	A	B	C	D	E	F	G
1	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}	X_{p1}
2	X_{p1}	X_{p1}	X_{p1}	X_{p3}	X_{p3}	X_{p3}	X_{p3}
3	X_{p1}	X_{p3}	X_{p3}	X_{p1}	X_{p1}	X_{p3}	X_{p3}
4	X_{p1}	X_{p3}	X_{p3}	X_{p3}	X_{p3}	X_{p1}	X_{p1}
5	X_{p3}	X_{p1}	X_{p3}	X_{p1}	X_{p3}	X_{p1}	X_{p3}
6	X_{p3}	X_{p1}	X_{p3}	X_{p3}	X_{p1}	X_{p3}	X_{p1}
7	X_{p3}	X_{p3}	X_{p1}	X_{p1}	X_{p3}	X_{p3}	X_{p1}
8	X_{p3}	X_{p3}	X_{p1}	X_{p3}	X_{p1}	X_{p1}	X_{p3}

Table 4 Example of crossover performed using $L_8(2^7)$ orthogonal table for X_{p2} and X_{p3}

Factor							
Exp.	A	B	C	D	E	F	G
1	X_{p2}	X_{p2}	X_{p2}	X_{p2}	X_{p2}	X_{p2}	X_{p2}
2	X_{p2}	X_{p2}	X_{p2}	X_{p3}	X_{p3}	X_{p3}	X_{p3}
3	X_{p2}	X_{p3}	X_{p3}	X_{p2}	X_{p2}	X_{p3}	X_{p3}
4	X_{p2}	X_{p3}	X_{p3}	X_{p3}	X_{p3}	X_{p2}	X_{p2}
5	X_{p3}	X_{p2}	X_{p3}	X_{p2}	X_{p3}	X_{p2}	X_{p3}
6	X_{p3}	X_{p2}	X_{p3}	X_{p3}	X_{p2}	X_{p3}	X_{p2}
7	X_{p3}	X_{p3}	X_{p2}	X_{p2}	X_{p3}	X_{p3}	X_{p2}
8	X_{p3}	X_{p3}	X_{p2}	X_{p3}	X_{p2}	X_{p2}	X_{p3}

where f is the factor index; j is the level index; $J_{i(avg)}$ is the average of the three results; $J_{i(std)}$ is the standard deviation of the three results; J_{Max} is the maximum average in the orthogonal experiment; J_{Maxstd} is the maximum standard deviation in the orthogonal experiment.

After the optimal level of all factors is selected, a new producer vector X_{npi} ($i = 1, 2, 3$) is generated. Finally, there are six producer vectors, X_{p1} , X_{p2} , X_{p3} , X_{np1} , X_{np2} , and X_{np3} , and the fitness values of these six producer vectors are calculated separately. The producer with the best fitness is the optimal producer X_{op} for the current iteration. Figure 1 shows the flowchart of the R-GSO. Next, update the position of producers, scroungers, and rangers according to the GSO mechanism.

3 Results and Discussions

In this paper, the order of the digital IIR filter remains constant and does not vary throughout the optimization process. The design target for the digital IIR filter involves minimizing the objective function described in equ. 6 while adhering to stability constraints outlined in equ. 7 and adhering to the design parameters listed in Table 1. This paper uses equ. 6 to evaluate the performance of LP, HP, BS, and BP filters and compare them with the hybrid Taguchi genetic algorithm (HTGA) [27], where the situation of the structures of IIR filters are specified in advance. For the LP filter, the weights v_1 , v_2 , v_3 , and v_4 in equ. 6 are set to 1, 1, 0.5, and 0.5, respectively.

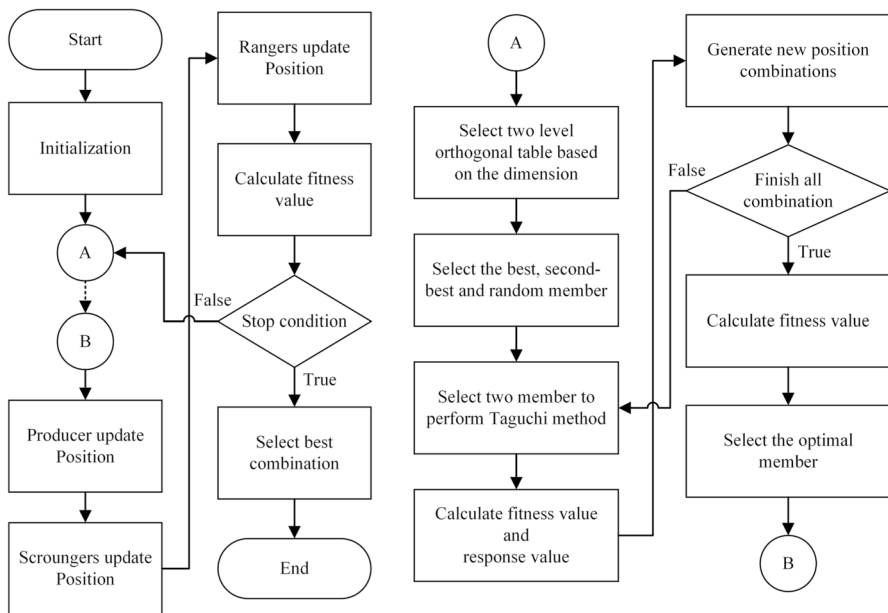


Fig. 1 The flowchart of the R-GSO algorithm. The algorithm starts with random initialization of the population members and calculates their fitness values. Then, the original producers with the best, second-best fitness scores and random members from the producer group are selected to perform the Taguchi method. The three producers will generate three crossover types. The Taguchi method constructs a two-level orthogonal table for each dimension to generate position combinations. Each new position combination is evaluated thrice, and the new member is selected based on equ. 8. Therefore, three new producers are generated. The optimal producer for the next iteration is determined from the highest fitness scores of these three new and original producers. Next, update the position of producers, scroungers, and rangers according to the GSO mechanism

For the HP filter, all of the weights v_1 , v_2 , v_3 , and v_4 are set to 1. For the BP and BS filters, the weights v_1 , v_2 , v_3 , and v_4 are set to 1, 1, 10, and 10, respectively. According to the filter specifications, the weights must be adjusted to design the filters.

3.1 Experimental Results of HTGA for Designing Digital IIR Filters

Under the situation of the structures of IIR filters are specified in advance, the HTGA parameters are set as follows: LP, HP filters dimension is 7, BS filter dimension is 9, BP filter dimension is 13, the search range is $[-2, 2]$, the population size is 100, the crossover rate is 0.8, the mutation rate is 0.1, and the generations is 500 [27]. Since the dimension of the LP and HP filters is 7, the orthogonal table $L_8(2^7)$ is chosen, while the orthogonal tables $L_{16}(2^9)$ and $L_{16}(2^{13})$ are selected for the BS and BP filters, respectively. The L_8 and L_{16} orthogonal tables have 8 and 16 experiments, respectively. According to the steps of the HTGA, if the number of mutations is not considered, the number of function evaluations is calculated as: population size + $(0.25 \times \text{population size} \times 0.8) \times \text{the number of Taguchi experiments} \times \text{generations}$

The parameters of digital IIR filters by using HTGA are as follows:

$$H_{LP}(z) = 0.044191 \frac{(z + 0.999891)(z^2 - 0.603517z + 0.985351)}{(z - 0.649020)(z^2 - 1.388808z + 0.736754)} \quad (9)$$

$$H_{HP}(z) = 0.058617 \frac{(z - 1.000535)(z^2 + 0.720845z + 0.989847)}{(z + 0.609721)(z^2 + 1.345146z + 0.726439)} \quad (10)$$

$$H_{BS}(z) = 0.423810 \frac{(z^2 + 0.431496z + 0.968503)(z^2 - 0.441200z + 1.012833)}{(z^2 + 0.849981z + 0.544481)(z^2 - 0.850355z + 0.546167)} \quad (11)$$

$$H_{BP}(z) = 0.027639 \frac{(z^2 + 0.092090z - 1.151911)(z^2 + 0.023145z - 1.028540)(z^2 + 0.119995z - 1.1024)}{(z^2 + 0.000155z + 0.4600)(z^2 - 0.643600z + 0.736177)(z^2 + 0.634888z + 0.733279)} \quad (12)$$

3.2 Experimental Results of R-GSO for Designing Digital IIR Filters

The R-GSO parameters are set as follows: LP, HP filters dimension is 7, BS filter dimension is 9, BP filter dimension is 13, the search range is $[-2, 2]$, the population size is 100, the maximum number of iterations is 500, the number of producer is 9, the percentage of rangers is 70%, the maximum search angle and turning angle of producer is π/α^2 , the longest search distance of producer is 6, and the initial angle is $\pi/6$. Since the dimension of the LP and HP filters is 7, the orthogonal table $L_8(2^7)$ is chosen, while the orthogonal tables $L_{16}(2^9)$ and $L_{16}(2^{13})$ are selected for the BS and BP filters, respectively. In the original GSO algorithm, the producer generates three search angles at the current position so that each additional producer number will have two more function evaluations. According to the steps of R-GSO, three Taguchi experiments are conducted. After each Taguchi experiment, the response values are calculated, and the best combination for that experiment is selected. Therefore, the number of function evaluations is calculated as: (population size + the number of producers $\times 2$) + (the number of Taguchi experiments $\times 3 + 3$) $\times$ iterations

Figures 2 and 3 shows the convergence curve of fitness value for the four types of digital IIR filters using the R-GSO and GSO algorithms, starting from the 50th iteration. It can be observed that the fitness value of the LP filter decreases rapidly in the early iterations and stabilizes after around 150 iterations. The fitness value of the HP filter is initially high, decreases significantly several times during the convergence process, and stabilizes after about 400 iterations. The fitness value of the BS filter gradually decreases with a moderate convergence rate, reaching stability after approximately 330 iterations. The fitness value of the BP filter still shows a slow downward trend at the end of 500 iterations, indicating that it has not fully converged. Compared with the original GSO algorithm, the R-GSO algorithm consistently exhibits lower final fitness values across all filter types. The R-GSO algorithm achieves faster convergence in the LP, BS, and BP filters. Moreover, in the HP filter, the R-GSO algorithm is able to escape local optima and converge to a better solution. Overall, the R-GSO algorithm is more efficient regarding convergence speed and more effective in avoiding local minima, as shown in figures 2 and 3. The parameters of digital IIR filters by using R-GSO are as follows:

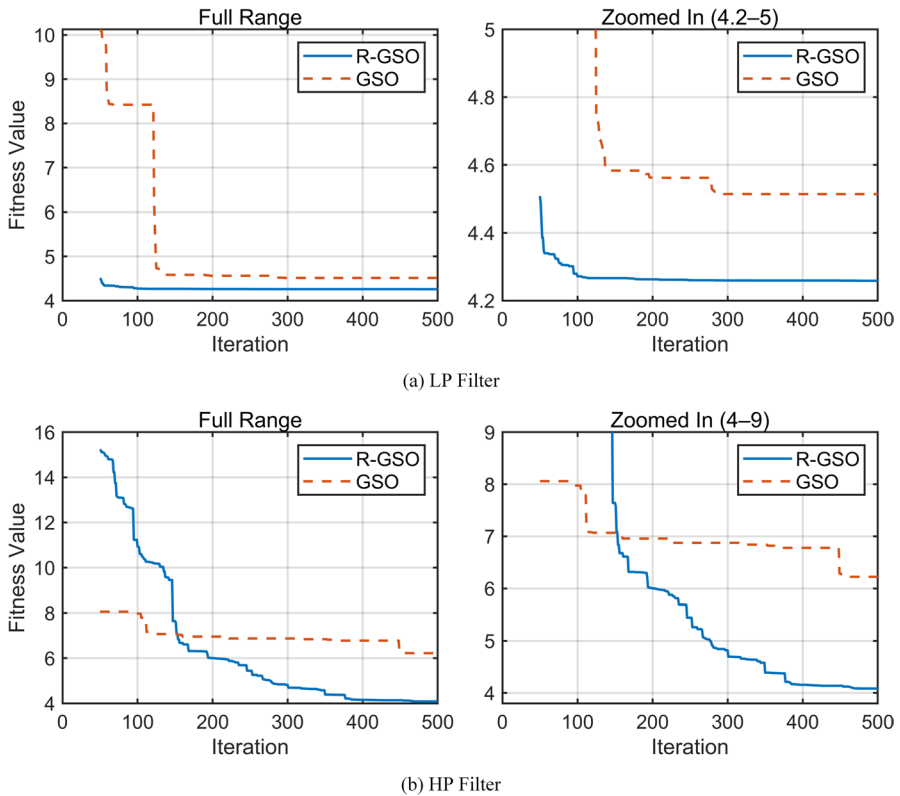


Fig. 2 The convergence curve of fitness value for **a** LP and **b**HP digital filters design using the R-GSO and GSO algorithms (starting from the 50th iteration)

$$H_{LP}(z) = 1.2413 \frac{(z + 0.9998)(z^2 - 0.2504z + 0.9943)}{(z - 0.6500)(z^2 - 1.3945z + 0.7141)} \quad (13)$$

$$H_{HP}(z) = 0.034 \frac{(z - 0.9187)(z^2 + 0.3521z + 1.0487)}{(z + 0.6656)(z^2 + 1.4122z + 0.7357)} \quad (14)$$

$$H_{BS}(z) = 0.4288 \frac{(z^2 - 0.3257z + 0.9612)(z^2 + 0.3280z + 0.9906)}{(z^2 + 0.7976z + 0.5091)(z^2 - 0.7993z + 0.5099)} \quad (15)$$

$$H_{BP}(z) = 0.0481 \frac{(z^2 + 0.2878z - 0.5896)(z^2 + 1.628z + 0.7151)(z^2 - 1.6237z + 0.8159)}{(z^2 + 0.6521z + 0.7553)(z^2 - 0.6034z + 0.7809)(z^2 + 0.0521z + 0.5326)} \quad (16)$$

3.3 Comparison and Discussion

Equations 2, 3, 4, 5, 6, 7, 8 and 9 is used to evaluate the parameters of the IIR filters designed by R-GSO, HTGA and GSO algorithm, as shown in Table 5. From the Table 5, it can be seen that the LP fitness value of R-GSO, HTGA and GSO algorithm

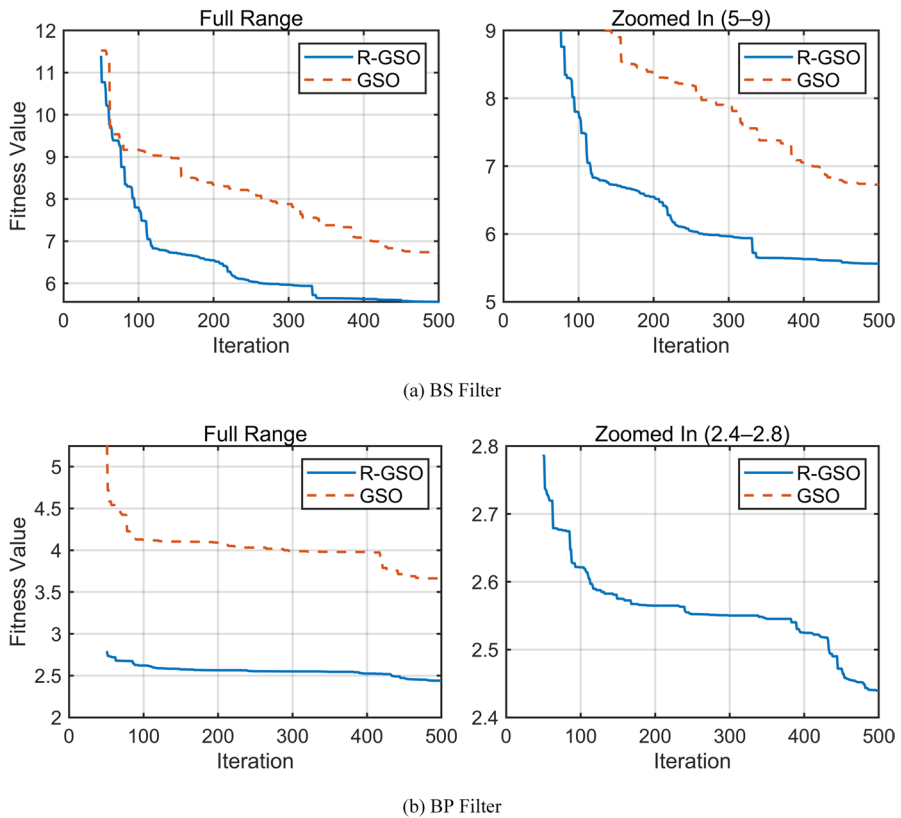


Fig. 3 The convergence curve of fitness value for **a** BS and **b** BP digital filters design using the R-GSO and GSO algorithms (starting from the 50th iteration)

Table 5 Comparison of R-GSO and HTGA Digital IIR Filter Experimental Results

Filter type				
Algorithm	LP	HP	BS	BP
R-GSO	4.2585	4.0842	5.5626	2.4395
GSO	4.5136	6.2238	6.7271	3.6634
HTGA	4.6478	5.4928	6.4821	3.1400

are 4.2585, 4.6478 and 4.5136, respectively; the HP fitness value of R-GSO, HTGA and GSO algorithm are 4.0842, 5.4928 and 6.2238, respectively; the BS fitness value of R-GSO, HTGA and GSO algorithm are 5.5626, 6.4821 and 6.7271, respectively; the BP fitness value of R-GSO, HTGA and GSO algorithm are 2.4395, 3.1400 and 3.6634, respectively. For the LP and HP filters, the number of function evaluations for R-GSO algorithm is approximately 72,500, while that for HTGA is about 130,000. For the BS and BP filters, the number of function evaluations for R-GSO algorithm is around 84,500, and that for HTGA is about 210,000.

In summary, considering the L_1 -norm approximation error, L_2 -norm approximation error, and the ripple sizes in the passband and stopband, the fitness values of

R-GSO algorithm are better than HTGA and GSO algorithm on LP, HP, BS, and BP filters. Regarding computational cost, the number of function evaluations required by R-GSO algorithm lies between HTGA and GSO algorithm.

4 Conclusion

This paper introduced an R-GSO algorithm that incorporates the GSO algorithm and Taguchi method to solve the problem of designing the optimal lowest-order serial digital IIR filters, where the structures of IIR filters are specified in advance. Unlike the original GSO algorithm with a single producer, the R-GSO algorithm selects a fixed number of multiple producers combined with the Taguchi method to determine the optimal producer in each iteration. The Taguchi method enhances the GSO algorithm's global search performance. This paper performed an R-GSO algorithm to solve the multi-parameter and multi-objective optimization problems of designing digital LP, HP, BP, and BS filters. Experimental outcomes indicate that the R-GSO algorithm proficiently optimizes digital LP, HP, BP, and BS filters. The R-GSO algorithm outperforms HTGA regarding both the fitness values and the number of function evaluations for LP, HP, and BS filters, thus providing a more efficient approach for optimizing these four types of filters and offering direct applicability in engineering scenarios that require high-performance digital filters. In this study, the R-GSO algorithm was compared only with the original GSO algorithm and HTGA, which effectively verified the improvements achieved by R-GSO. Further comparisons with other recently developed advanced metaheuristic algorithms would provide a more comprehensive assessment of the proposed algorithm's performance for future work.

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Declarations

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